Band-1 Hadamard Order 7988 – Miyamoto (Paley-II Seed H_1996 q=997 + C_1998); Miyamoto-only NOT in SPK; Gap 2052 Still OPEN

H_1996 Paley-II PASS; C_1998 PASS; H_7988 Miyamoto PASS; classical SPK FAIL -> Miyamoto-only NOT in SPK; gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

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Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list * Human+AI attack order * ZoraASI mission-control.
Parents. `2026-09-16_Band1_Hadamard_7732_Miyamoto`; `2026-09-16_Band1_Hadamard_7604_PaleyI`.
Lab. `/workspace/experiments/band1/hadamard_7988_miyamoto/`.
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Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 1996 and 7988 only** (verified $\max|HH^\top - nI| = 0$), via Miyamoto with Paley-II seed H_{1996} = Paley-II($q = 997$) and symmetric Paley conference C_{1998} over GF(1997) ($q = 1997 \equiv 1 \pmod 4$). Classical SPK for 7988: Paley-I **FAIL**s ($q = 7987 = 7^2 \cdot 163$ composite); Paley-II **FAIL**s ($q = 3993 = 3 \cdot 11^3$ NOT prime power); Kronecker **FAIL**s (4×1997). Therefore order **7988 is a Miyamoto-only bonus NOT in SPK**. It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under SPK. `clay_closed = false`. Prior ship: **7732 Miyamoto-only NOT in SPK** (seed **1932** IN SPK). Prior SPK seeds **1932 / 7604 / 1876+7508 / 1872 / 1860 / 7444 / 1848 / 7396 / 1788 / 888 / 1776 / 860 / 1720 / 6884 / 1708 / 840 / 1680 / 6724 / ...** retained; Miyamoto-only bonuses **7732/7492/7156/7108/6836/6788/6676/6484/6404/6388/6196/...** and **7988** excluded; **1996** folded into ONBOX/SPK when verified.

Abstract

Finite constructions of orders **1996** and **7988**.

- H_{1996} = Paley-II ($q = 997$) – preferred seed (Paley-I $q = 1995$ FAIL; Double FAIL; Kronecker 4×499 FAIL)
- C_{1998} = Paley conference over GF(1997), $q = 1997$
- H_{7988} = Miyamoto(H_{1996}, C_{1998})
- Paley-I for 7988: **SKIP** ($q = 7987$ composite)
- Paley-II for 7988: **SKIP** ($q = 3993$ NOT prime power)
- Kronecker for 7988: **SKIP** ($4 \cdot 1997$ not admissible)

SPK: 7988 Miyamoto-only NOT in SPK. Seed 1996 enters SPK (Paley-II). Prior SPK **1932 / 7604 / 7508 / 1876 / ...** retained.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

`python3 verify_hadamard_7988.py`

Results. overall_pass=true; units 130/130; wall 417.2427s; all max|HHˆT-nI|=0.

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Arithmetic

- Target $n = 7988 = 4 \cdot 1997$; 1997 prime $\equiv 1 \pmod 4$ -> Miyamoto-eligible.
- Need H_{1996} and C_{1998} (Paley conference over prime field $q = 1997$).
- Seed: Paley-II $q = 997$ (prime $\equiv 1 \pmod 4$) -> H_{1996} (Paley-I $q = 1995 = 3 \cdot 5 \cdot 7 \cdot 19$ FAIL; Double FAIL; Kronecker 4×499 FAIL).
- SPK: Paley-I $q = 7987$ **FAIL**; Paley-II $q = 3993$ **FAIL** (NOT prime power); Kronecker **FAIL** -> **Miyamoto-only NOT in SPK**.
- Prior: 7732 Miyamoto-only shipped; this ship $q = 1997 \rightarrow n = 7988$.

Constructions (verified)

Object	Method	max|HHˆT-nI|	Wall (s)
H_1996 Paley-II	Paley-II q=997	0	4.2721
C_1998	Paley conf GF(1997) q=1997	0	5.4132
H_7988 Miyamoto	Miyamoto(H_1996 Paley-II, C_1998)	0	403.2682
H_7988 Paley-I	SKIP q=7987=7^2*163	–	–
H_7988 Paley-II	SKIP q=3993=3*11^3	–	–
H_7988 Kronecker	SKIP 4*1997 not admissible	–	–

SPK closure note

Seeds include prior SPK ships through **7732** lineage (**1932** Paley-I; **7604** Paley-I; **1876/7508**; **1872**; **1860/7444**; **1848/7396**; **1788**; **888/1776**; **860/1720/6884**; **1708**; **6724**; **840/1680**; **...**; **NOT** 7732/7492/7156/7108/6836/6788/6676/6484/6404/6388/6196) + verified **1996** (Paley-II); **exclude** Miyamoto-only bonuses including **7988**. Known count 1592. Next constructive gap **2052** (OPEN).

Quantity	Value
7988 in SPK	**False** (Miyamoto-only)
1996 in SPK	**True** (Paley-II q=997)
1932 / 7604 / 7508 / 1876 in SPK	**True** (prior)
7732 / 7492 / 7156 in SPK	**False** (Miyamoto-only; excluded)

Honesty gates

- `clay_closed = false`
- GATE-04 collection **LOCKED**; no invented J
- Finite-order constructions **1996** and **7988** ONLY (verified max|HHT-nI|=0)
- **NO** full Hadamard conjecture claim; **NO** finished-ToE; **NO** RH / P!=NP / YM / Frankl / NS-solved / AC-solved
- SPK membership only via classical Paley-I / Paley-II / Kronecker (**Miyamoto alone is NOT SPK**)
- Standard constructive gap **2052** remains **OPEN**

Next optional

- **8068 Miyamoto Kronecker(4x504)** OR **8116 Miyamoto PaleyI(q=2027)** OR method-attack SPK gap **2052** OR Track B / Object 1 meter await